Spatially Guided LLMs for Evaluating Operational Design Domain Compliance in Autonomous Driving

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Can you drive a Tesla in the river?

Problem

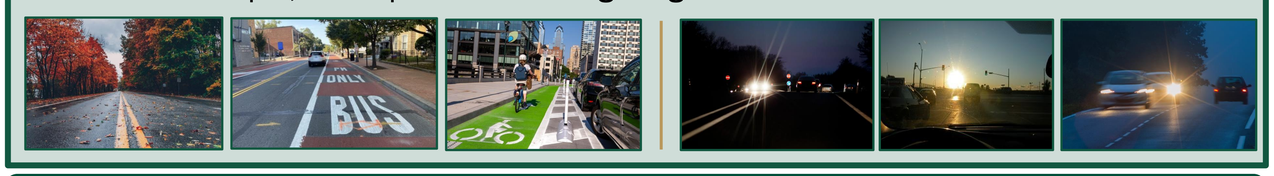
Driving Automation Systems (DAS) are designed to operate only within a defined **Operational Design Domain (ODD)**, which specifies the environmental, roadway, and traffic conditions under which the system is intended to function safely.

However, in practice:

- ODDs are written in **informal** natural language.
- They lack **precision** and **formal** structure.
- There is no scalable mechanism to verify **dataset compliance**.
- There is no systematic method to **refine ODD specifications.**

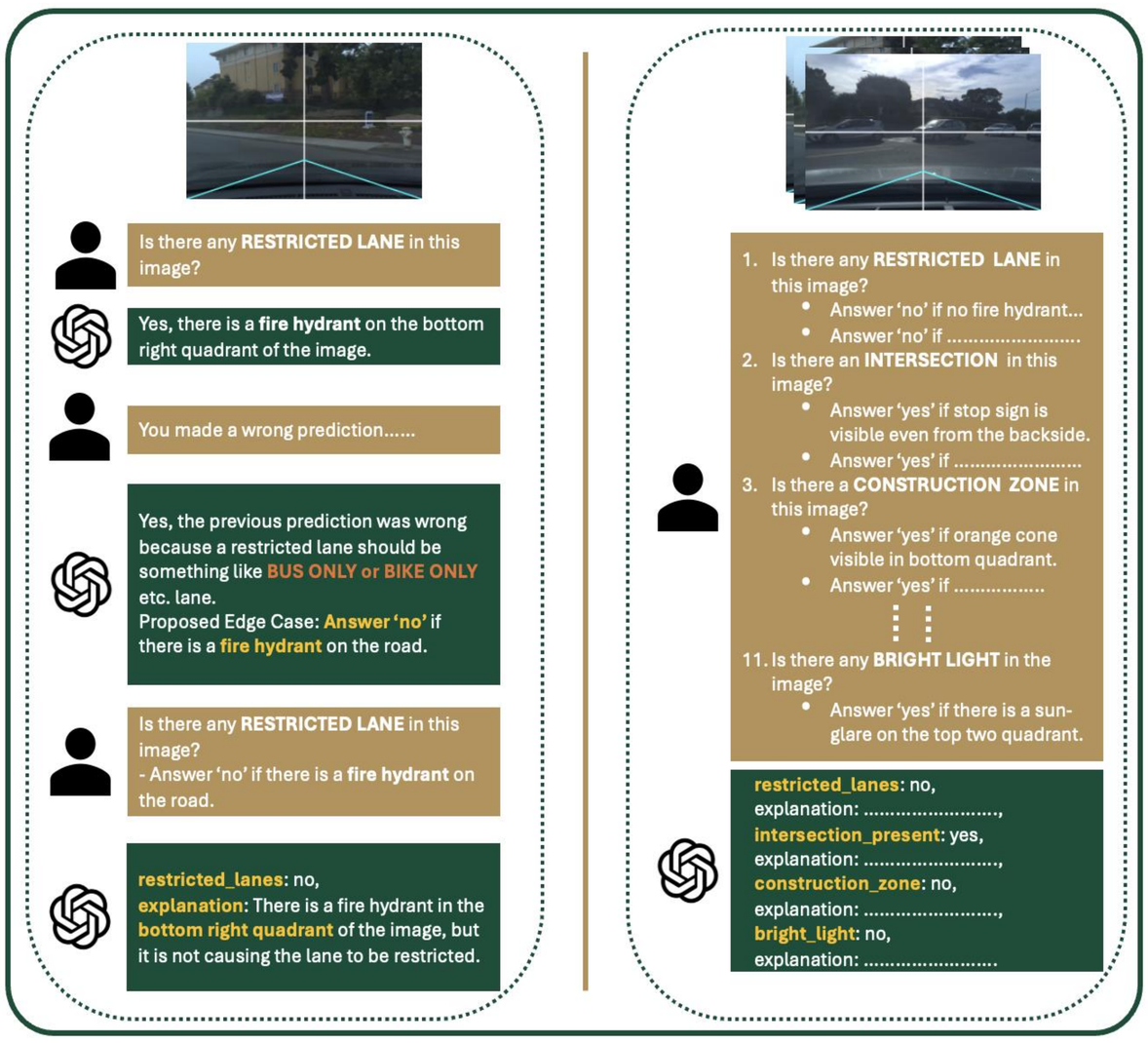
As a result:

- Driving datasets may include situations, the system **was not designed to handle.**
- ODD violations can silently **leak** into development pipelines.
- Valuable in-ODD data may remain **unused**.
- Manual compliance checking is **labor-intensive** and inherently subjective. For example, what qualifies as **"bright light"?**

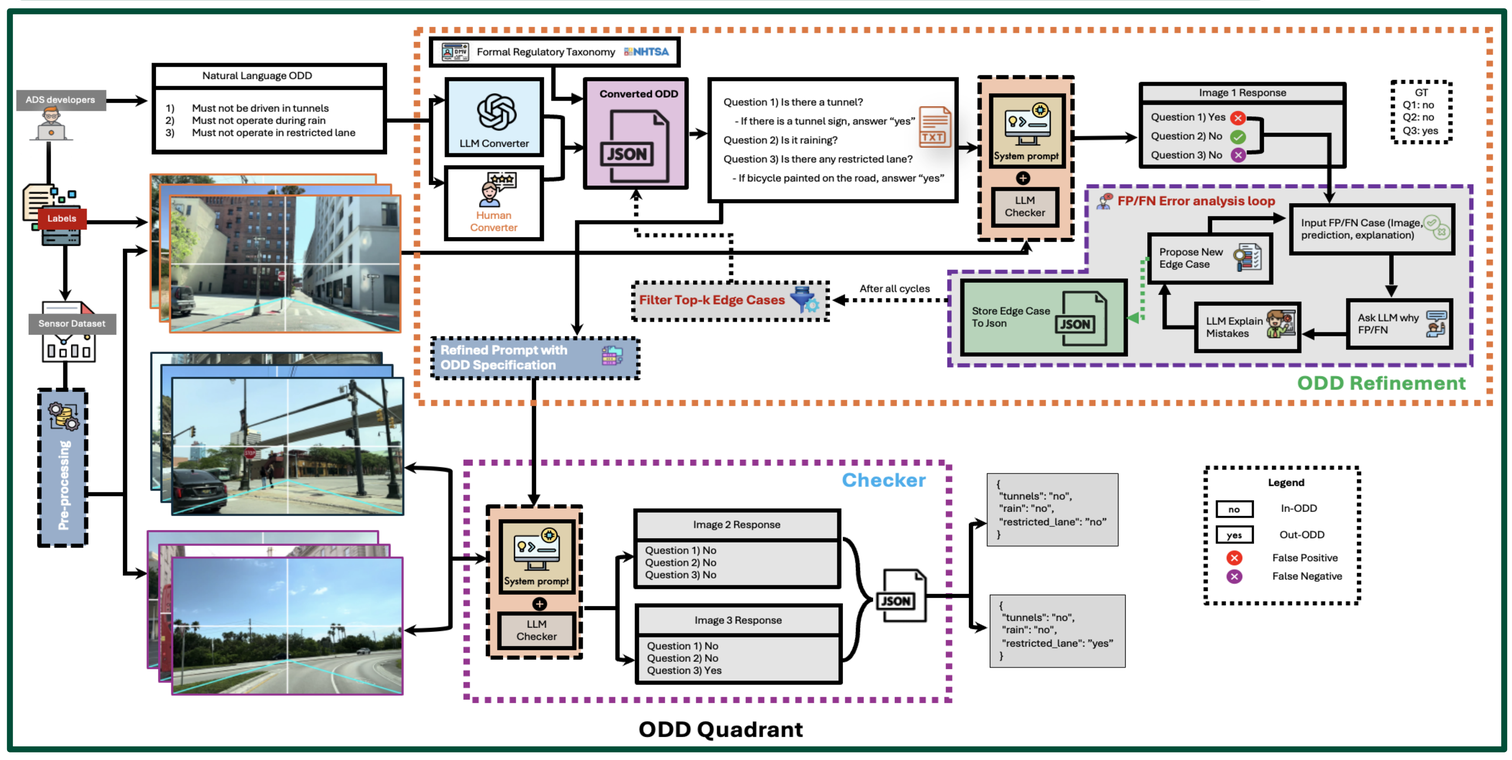


ODD-Quadrant : A structured framework that uses LLMs to turn unclear ODD descriptions into clear yes-or-no questions. Instead of analyzing the whole scene at once, we divide each driving image into **four regions** and a forward **V-shaped** area that represents the vehicle's path. This helps the model focus on where **risks actually matter**. By **refining the prompts** over time, the system becomes better at identifying **rare and complex** edge cases. As a result, it can detect ODD violations more accurately and efficiently across both **seen** and **new datasets**.

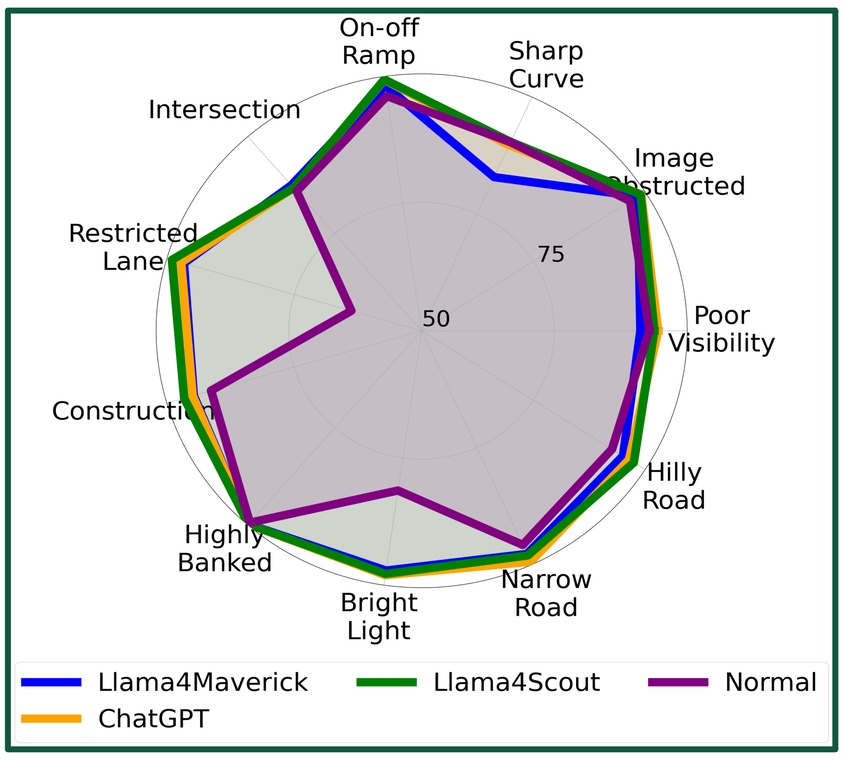
Contribution



How ODD-Quadrant Works?

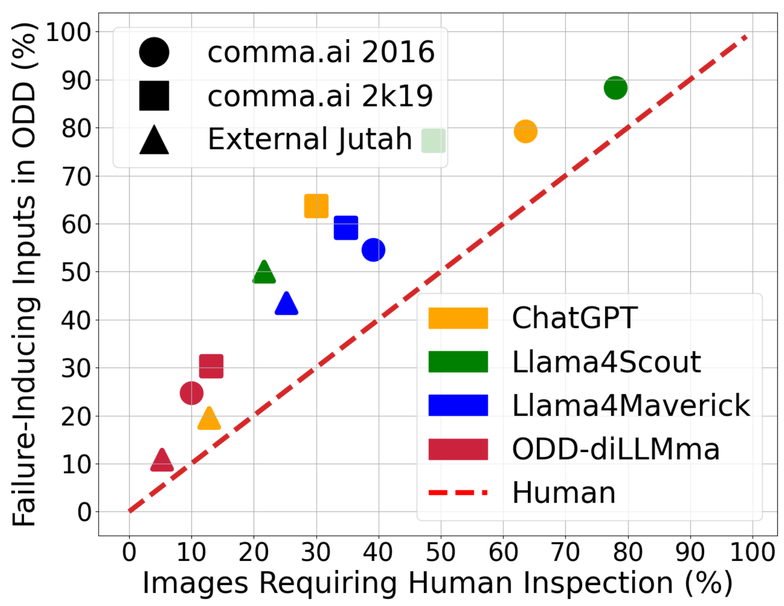


Semantic Dimensions Accuracy

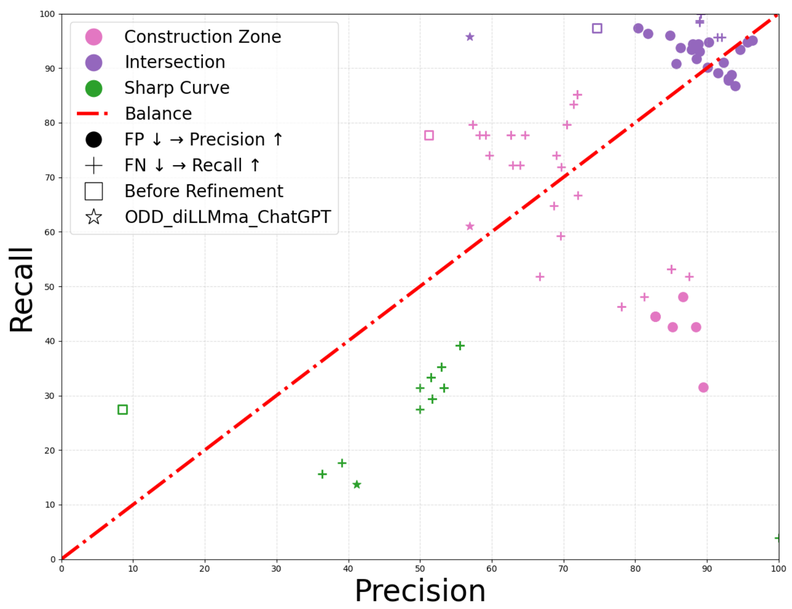


Results

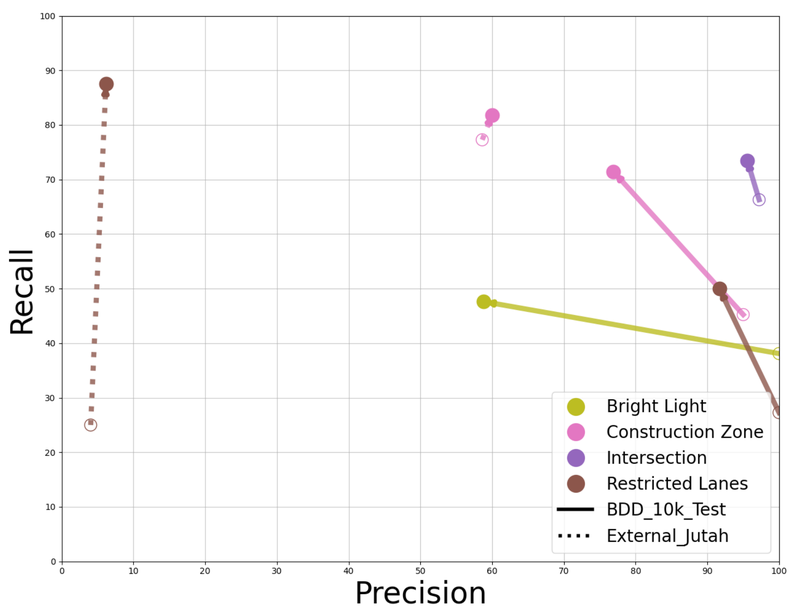
LLMs detect more ODD failures with less human effort than manual review



Refinement improves both precision or recall



Our method improves recall on unseen data, demonstrating strong generalization



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